12.25 The Largest of Many

WORKSHEET

The question

STEP III 2025, Question 11

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself The largest of $X_1, \dots, X_n$ is at most y exactly when what happens?

STAGE 2 The quantiles

Ask yourself Solve $\mathrm{G}(L) = \alpha$ for L. Then solve $\mathrm{G}(U) = 1 - \alpha$.

STAGE 3 Large n

Ask yourself Write $\alpha^{1/n} = e^{(\ln \alpha)/n}$. For large n the exponent is small. What is $1 - \alpha^{1/n}$ approximately?

STAGE 5 A 90% interval

Ask yourself Take $\alpha = 0.05$. What are $\ln \frac{1}{\alpha} = \ln 20$ and $\ln \frac{1}{0.95}$, approximately?

Try this next

1. **The expected maximum.** Show that $\mathrm{E}(Y) = \frac{1}{\lambda}(1 + \frac{1}{2} + \cdots + \frac{1}{n})$. (*Hint:* by memorylessness, after the first of the n lifetimes ends, the remaining $n - 1$ start afresh.)

2. **The minimum.** Show that the smallest of $X_1, \dots, X_n$ is exponential with parameter $n\lambda$.

3. **The median exactly.** Find the exact median of Y and compare it with the approximation $\frac{1}{\lambda}(\ln n - \ln \ln 2)$ for $n = 10$.

4. **Uniform maxima.** If $X_1, \dots, X_n$ are uniform on $[0, 1]$, find the distribution of their maximum M, and show that $n(1 - M)$ is approximately exponential for large n. Which of the three extreme-value shapes is this?

Further reading

- **Stuart Coles, *An Introduction to Statistical Modeling of Extreme Values*** (Springer). The standard introduction, with real data on floods and sea levels.
- **Emil Julius Gumbel, *Statistics of Extremes*** (Dover). The classic.
- **David Spiegelhalter, *The Art of Uncertainty*** (Pelican), on rare events and how to think

about them.

The full guided solution, and what the question leads to at university, is in Section 12.25:
https://beyondstep.keithmcnulty.org/ch12/extremes.html